

LUKE YUEN

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EDUCATION

Stanford University School of Engineering - Stanford, CA | M.S., Mechanical Engineering, Expected 2027 | GPA 4.0/4.0
Relevant Coursework: Intro to Robotics, Principles of Robot Autonomy, Collaborative Robotics

Rice University - Houston, TX | B.S., Mechanical Engineering, 2025, GPA 3.9/4.0

SKILLS

CAD & PLM: Siemens NX, Teamcenter, SolidWorks, Fusion 360, GD&T, Tolerance Stack-Up Analysis, Over-molding

Manufacturing & DFM: DFM/DFA for high-volume production, Injection Molding, Fixture and Tooling Design, Engineering Drawings, BOM Management, First-Article Inspection

Prototyping & Testing: Stratasys PolyJet, FDM/SLA 3D Printing, Silicone Casting, Functional Verification, Instrumentation, Test Documentation, Assembly SOPs

EXPERIENCE

Apple - Culver City, CA | *Product Design Engineer Intern, Beats* | Mar 2026 - Present

- Support confidential product development via mechanical design, tight-tolerance packaging, and architecture iteration
- Utilize Siemens NX for geometric modeling and cross-functional layout integration with electrical/acoustic teams
- Partner with external vendors to manage component prototyping, tooling reviews, and manufacturing line readiness

Stealth eVTOL Drone Startup - San Francisco, CA | *Mechanical Design Engineer* | Jul 2025 - Sep 2025

- Designed parametric SolidWorks CAD for airframe subassemblies and propulsion mounts, emphasizing structural stiffness, serviceability, and assembly
- Prepared STEP models and manufacturing drawing updates for vendor communication, packaging reviews, and initial prototype BOM
- Executed preliminary CFD on body concepts, advancing two configurations based on aerodynamic performance and manufacturability constraints

Carbon Robotics - Everett, WA | *Mechanical Engineering Intern* | May 2025 - Jul 2025

- Produced SolidWorks drawings and executed GD&T tolerance stacks on complex assemblies to eliminate stack-ups while ensuring tool clearance
- Released four vendor-ready drawing packages and authored a 20-page assembly SOP to support first-article builds and manufacturing line bring-up
- Troubleshoot production-line modules end-to-end, updating mechanical components, laser calibrations, and power settings before field deployment

VacuTrac - Integrated Retraction & Suction Device - Houston, TX | *Mechanical Engineering Lead* | Aug 2024 - May 2025

- Designed a toolless suction tip interface that reduced bench-tested operating room changeover time by two minutes and streamlined user workflows
- Consolidated five subassemblies into two and eliminated four unique fasteners to significantly reduce part count, assembly time, and bill of materials cost

MD Anderson IDEAS Lab - Houston, TX | *Engineering Intern* | May 2024 - Jan 2025

- Fabricated SLA/FDM prototypes and silicone-cast handles, translating surgeon feedback into six rapid CAD revisions to optimize ergonomics
- Converted insights from 15 clinical sessions into 20 strict design requirements guiding subsequent manufacturing and sterilization loops

Rice Computational Imaging Lab - Houston, TX | *Researcher* | Jul 2023 - Jan 2025

- Packaged an enclosure and strap assembly for a wearable sensor device to maintain a ± 5 mm placement window across diverse body types

PUBLICATIONS & AWARDS

Co-author, *IEEE Pulse* (2025): "Novel, Integrated Tissue Retraction and Suction Device for Spinal Surgery"

National Inventors Hall of Fame Collegiate Inventors Competition Finalist (2025), Shapiro Showcase 1st (2025), Huff OEDK 2nd (2025), VentureWell DEBUT Honorable Mention (2025)